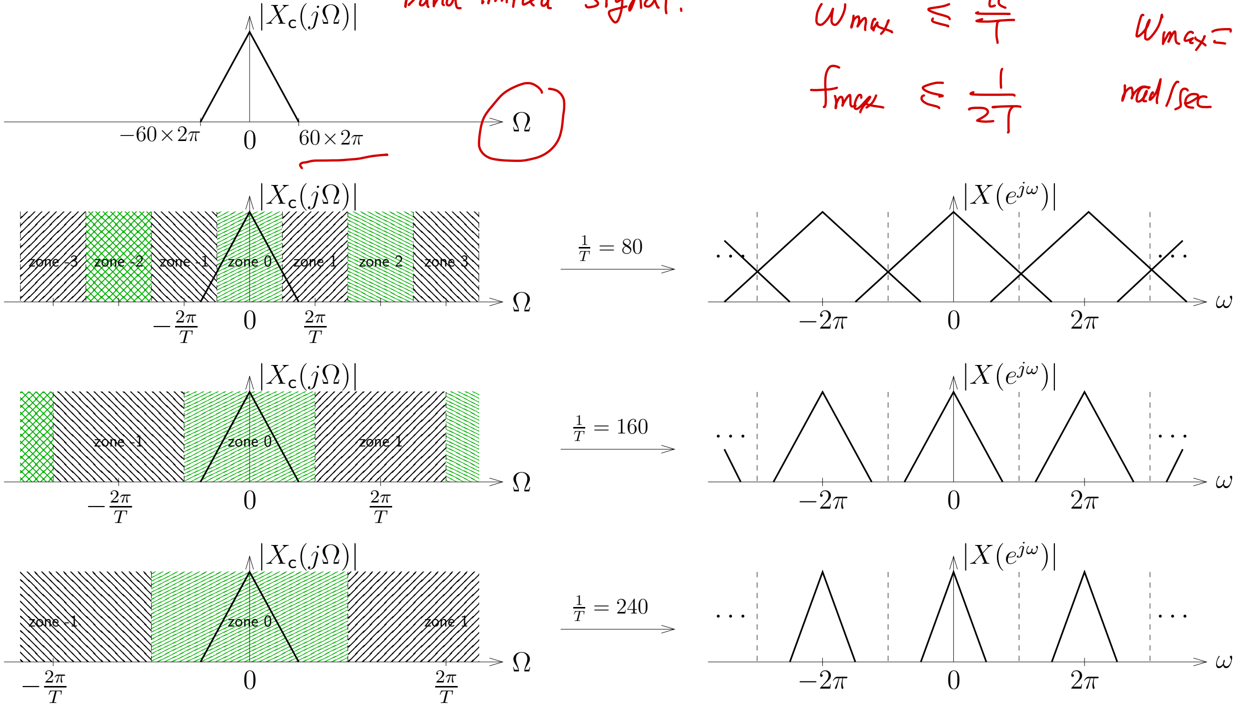
Nyquist sampling – example:

- Suppose that $X_c(j\Omega)$ has the CTFT magnitude sketched below, where $|f|_{\max} = 60$ Hz.
- Nyquist sampling specifies that $\frac{1}{T} > 2 \times 60 = 120\,\text{Hz}$.
- The figures below illustrate the Nyquist zones and the resulting DTFT for several choices of sampling rate $\frac{1}{T}$.

band limited signal.

$\omega_{max} \leq \frac{\pi}{T}$ $\quad$ $\omega_{max} = 2\pi f_{max}$

$f_{max} \leq \frac{1}{2T}$ $\quad$ rad/sec $\quad$ Hz

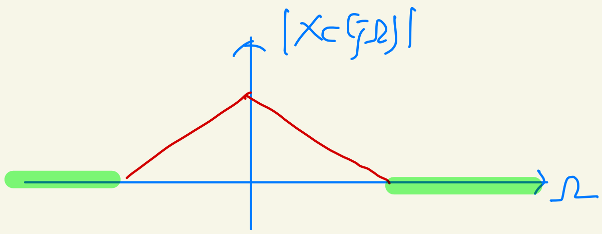


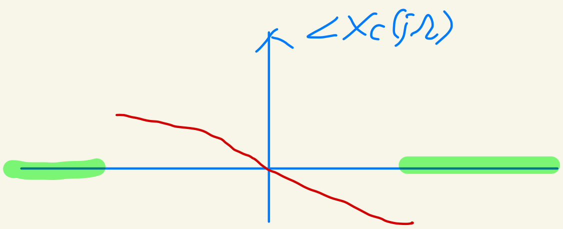
- Note the presence of aliasing in the DTFT when $\frac{1}{T}$ is less than the Nyquist sampling rate (i.e., $2|f|_{\max}$ Hz).

$$X_c(j\Omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\Omega t}\, dt$$

In general, $X_c(j\Omega)$ is complex-valued

$$X_c(j\Omega) = |X_c(j\Omega)|\, e^{j\angle X_c(j\Omega)}$$





$z \in \mathbb{C}$

$z = re^{j\theta}$

$\boxed{r = 0 \implies z = 0}$

$\cancel{\theta = 0 \implies z = 0\ ?}$

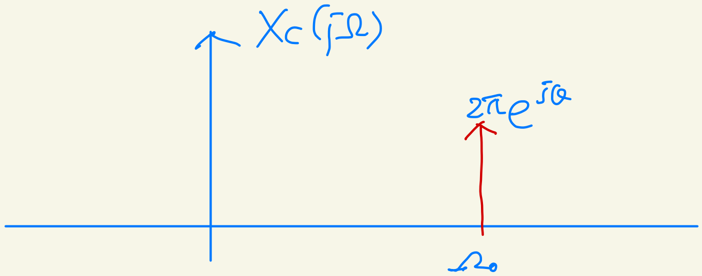
$$\text{support}\left(X_c(j\Omega)\right) = \{\Omega : X_c(j\Omega) \neq 0\}$$

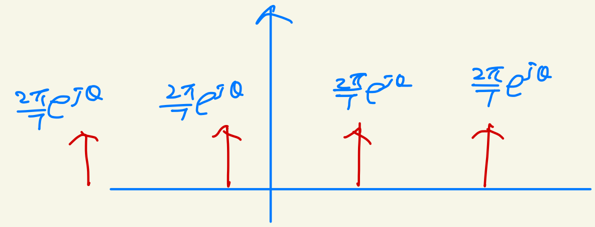
$$\|$$

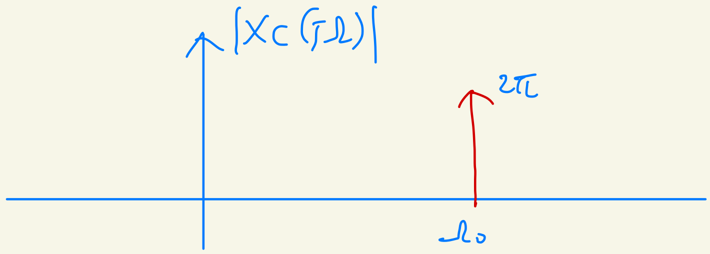
$$\text{support}\left(|X_c(j\Omega)|\right)$$

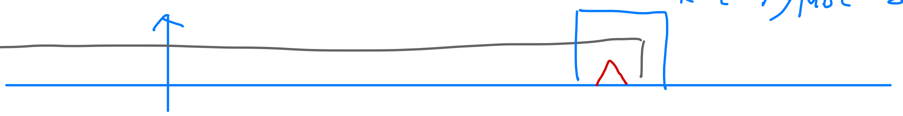
$$x(t) = e^{j(\Omega_0 t + \theta)} = e^{j\theta} e^{j\Omega_0 t}$$

$$X_c(j\Omega) = e^{j\theta}\, 2\pi\, \delta(\Omega - \Omega_0)$$





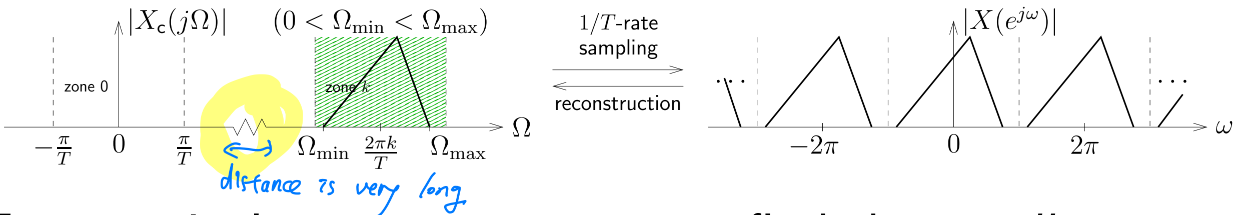




kth Nyquist zone with smaller width $\frac{2\pi}{T}$

Bandpass sampling – complex-valued signals:

- Consider a complex-valued bandpass signal of the form



- For practical reasons, we want to find the *smallest* possible sampling freq $\frac{1}{T}$. Then how do we choose k?

$\frac{2\pi k-\pi}{T} < 2\pi f_{min}$

- First, note that fitting into the k^{th} Nyquist zone requires

capture the spectrum within the kth Nyquist zone.

$$\left\{\begin{array}{l} \frac{2\pi k-\pi}{T} < \Omega_{\min} \\ \Omega_{\max} \le \frac{2\pi k+\pi}{T} \end{array}\right\} \overset{\Omega=2\pi f}{\Longleftrightarrow} \quad \frac{k-0.5}{f_{\min}} < T \le \frac{k+0.5}{f_{\max}}.$$

$\Omega_{min} = 2\pi f_{min}$ $\quad \Omega_{max} = 2\pi f_{max}$

- Also, note that T increases ($\frac{1}{T}$ decreases) as k increases. Thus, we want to find the largest integer k that satisfies

$$\frac{k-0.5}{f_{\min}} < \frac{k+0.5}{f_{\max}},$$

feasibility

and then use this "$k_{\max}$" to set the sampling interval as

$$T = \frac{k_{\max}+0.5}{f_{\max}}.$$

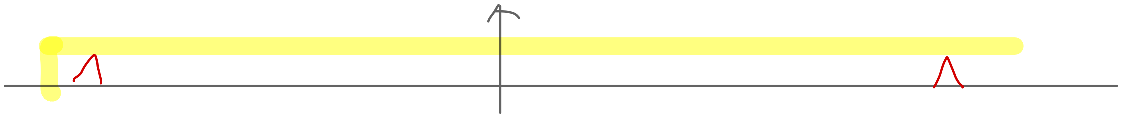
$f_{max} k - 0.5 f_{max} < f_{min} k + 0.5 f_{min}$

$k \le \left\lceil \frac{f_{min} + f_{max}}{2(f_{max} - f_{min})} \right\rceil - 1$

- After a bit of algebra, one can find

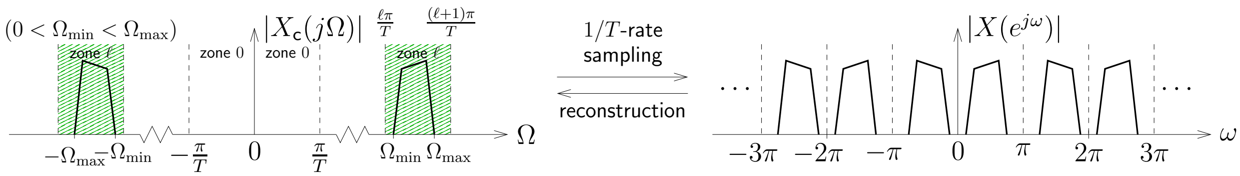
$$\frac{1}{T}\bigg|_{\min} = \frac{f_{\max}}{\left\lceil \frac{1}{2}\frac{f_{\max}+f_{\min}}{f_{\max}-f_{\min}} \right\rceil - \frac{1}{2}}$$

where $\lceil \cdot \rceil$ denotes the "ceiling" operation, i.e., rounding up to the nearest integer.



Bandpass sampling – real-valued signals:

- Consider a real-valued bandpass signal of the form



Want "half-spectrum" to fit into a "half-Nyquist" zone. These zones are width-$\frac{\pi}{T}$ and mirrored across $\Omega = 0$.

- Again, want to find smallest possible sampling freq $\frac{1}{T}$.

- Fitting into the ℓ^{th} half-Nyquist zone implies

$$\left\{ \begin{array}{l} \frac{\ell\pi}{T} < \Omega_{\min} \\ \Omega_{\max} \le \frac{(\ell+1)\pi}{T} \end{array} \right\} \quad \Leftrightarrow \quad \frac{\ell}{2f_{\min}} < T \le \frac{\ell+1}{2f_{\max}}$$

- Note that T increases ($\frac{1}{T}$ decreases) as ℓ increases. Thus, we want to find the largest integer ℓ that satisfies

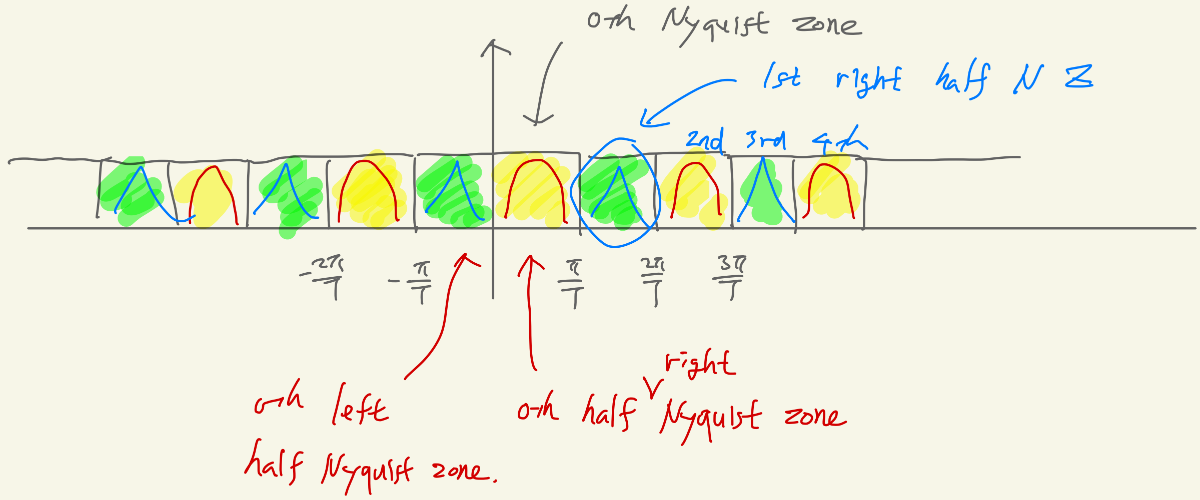
$$\frac{\ell}{2f_{\min}} < \frac{\ell+1}{2f_{\max}},$$

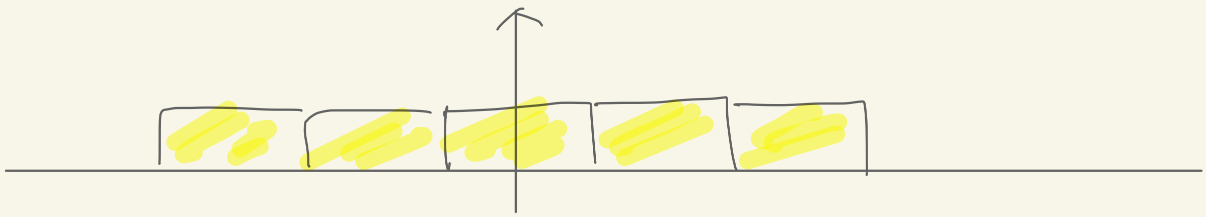
and then use this "$\ell_{\max}$" to set the sampling interval as

$$T = \frac{\ell_{\max}+1}{2f_{\max}}.$$

- After a bit of algebra, one can find

$$\frac{1}{T}\bigg|_{\min} = \frac{2f_{\max}}{\left\lceil \frac{f_{\min}}{f_{\max}-f_{\min}} \right\rceil}$$

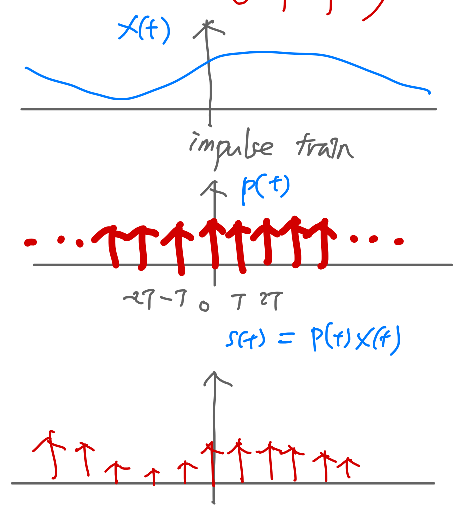

0th Nyquist zone
1st right half N Z
2nd 3rd 4th
−2π/T
−π/T
π/T
2π/T
3π/T
0th left half Nyquist zone.
0th half right Nyquist zone



Sampling — DTFT/CTFT relationship:

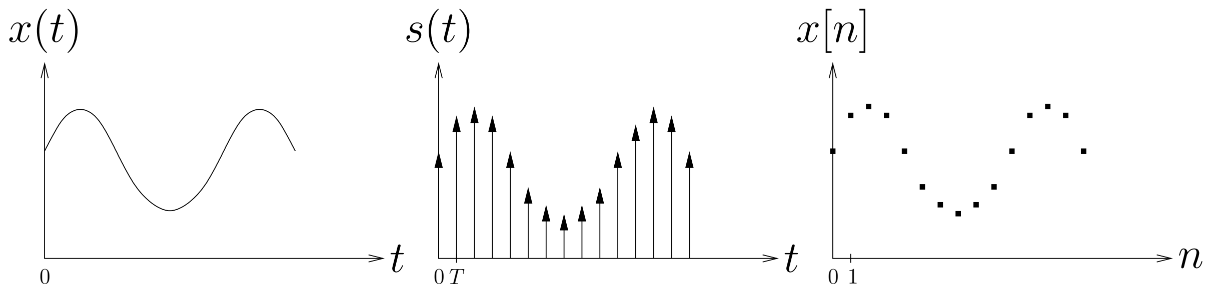
To fully understand aliasing, we need to derive the relationship between the CTFT $X_c(j\Omega)$ and the DTFT $X(e^{j\omega})$.

$$\begin{aligned}
X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \\
&= \sum_{n=-\infty}^{\infty} \underbrace{x(nT)}_{\text{function evaluation of } x(t) \text{ at } t=nT} e^{-j\omega n} \qquad x[n] = x(nT) \\
&= \sum_{n=-\infty}^{\infty} \left(\int_{-\infty}^{\infty} x(t)\delta(t-nT)dt\right) e^{-j\omega n} \qquad \text{sifting property of } \delta \\
&= \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} \underbrace{x(t)\delta(t-nT)e^{-j\omega n}}_{\text{nonzero iff } n = t/T.} dt \\
&= \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} x(t)\delta(t-nT)e^{-j\omega \frac{t}{T}} dt \\
&= \int_{-\infty}^{\infty} \underbrace{x(t) \sum_{n=-\infty}^{\infty} \delta(t-nT)}_{\triangleq\, s(t)} e^{-j\frac{\omega}{T}t} dt \qquad \text{comb function} \\
&= S_c\left(j\frac{\omega}{T}\right)
\end{aligned}$$



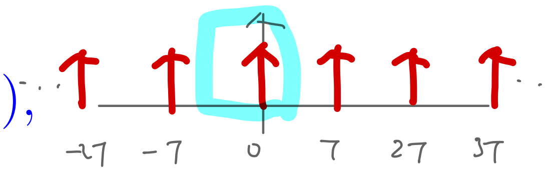
where we call $s(t)$ the "continuous-time sampled signal."

Sampling — DTFT/CTFT relationship (cont):



Notice that $s(t) = x(t)p(t)$ for the T-periodic "pulse train"

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT),$$



which has the Fourier series representation

T-periodic

$s(t) = p(t)\,x(t)$

$= \frac{1}{T} \sum_{k=-\infty}^{\infty} e^{j\frac{2\pi}{T}kt} x(t)$

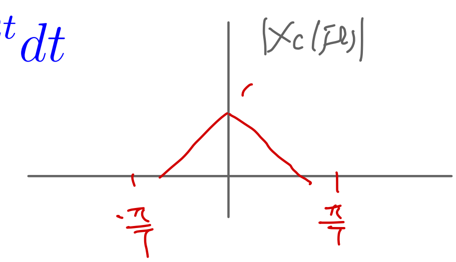
$$p(t) = \frac{1}{T} \sum_{k=-\infty}^{\infty} e^{j\frac{2\pi}{T}kt},$$

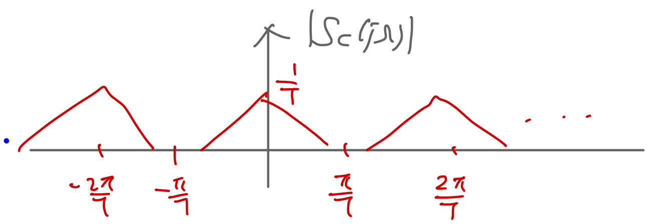
$p(t) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{T}kt}$

$a_k = \frac{1}{T}\int_{-T/2}^{T/2} p(t) e^{-j\frac{2\pi}{T}kt}\,dt$

and thus

$$\begin{aligned}
S_c(j\Omega) &= \mathcal{F}_{\text{CTFT}}\{s(t)\} = \mathcal{F}_{\text{CTFT}}\{x(t)p(t)\} \\
&= \int_{-\infty}^{\infty} \left(x(t) \frac{1}{T} \sum_{k=-\infty}^{\infty} e^{j\frac{2\pi}{T}kt} \right) e^{-j\Omega t} dt \\
&= \frac{1}{T} \sum_{k=-\infty}^{\infty} \int_{-\infty}^{\infty} x(t) e^{-j(\Omega - \frac{2\pi}{T}k)t} dt \\
&= \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\big(j(\Omega - \tfrac{2\pi}{T}k)\big).
\end{aligned}$$

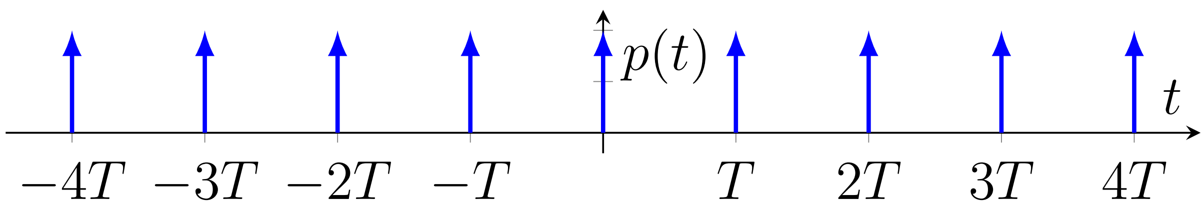




Sampling — DTFT/CTFT relationship (cont):

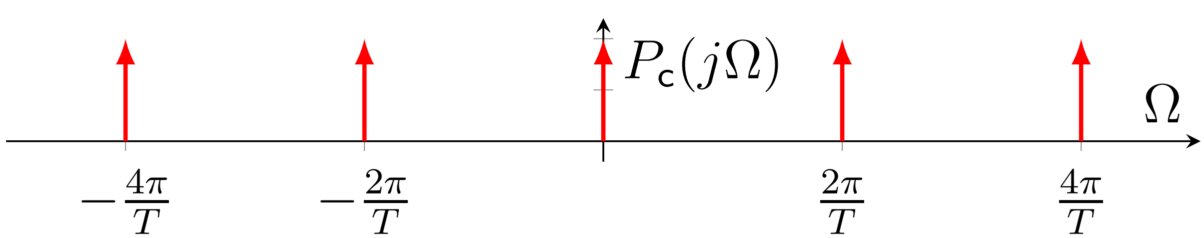
Alternative approach: Recall that the T-periodic pulse train, a.k.a. comb function, is given by

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT).$$



Since $p(t)$ is T-periodic, it is written as a Fourier series, from which its Fourier transform is derived as follows:

$$p(t) = \frac{1}{T} \sum_{k=-\infty}^{\infty} e^{j\frac{2\pi}{T}kt} \overset{\text{FT}}{\longleftrightarrow} \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta\left(\Omega - \frac{2\pi k}{T}\right).$$



By the multiplication property, we obtain

$$x(t)p(t) \overset{\text{FT}}{\longleftrightarrow} \frac{1}{T} \sum_{k=-\infty}^{\infty} X_{\mathsf{c}}\left(\Omega - \frac{2\pi k}{T}\right).$$

Sampling — DTFT/CTFT relationship (cont):

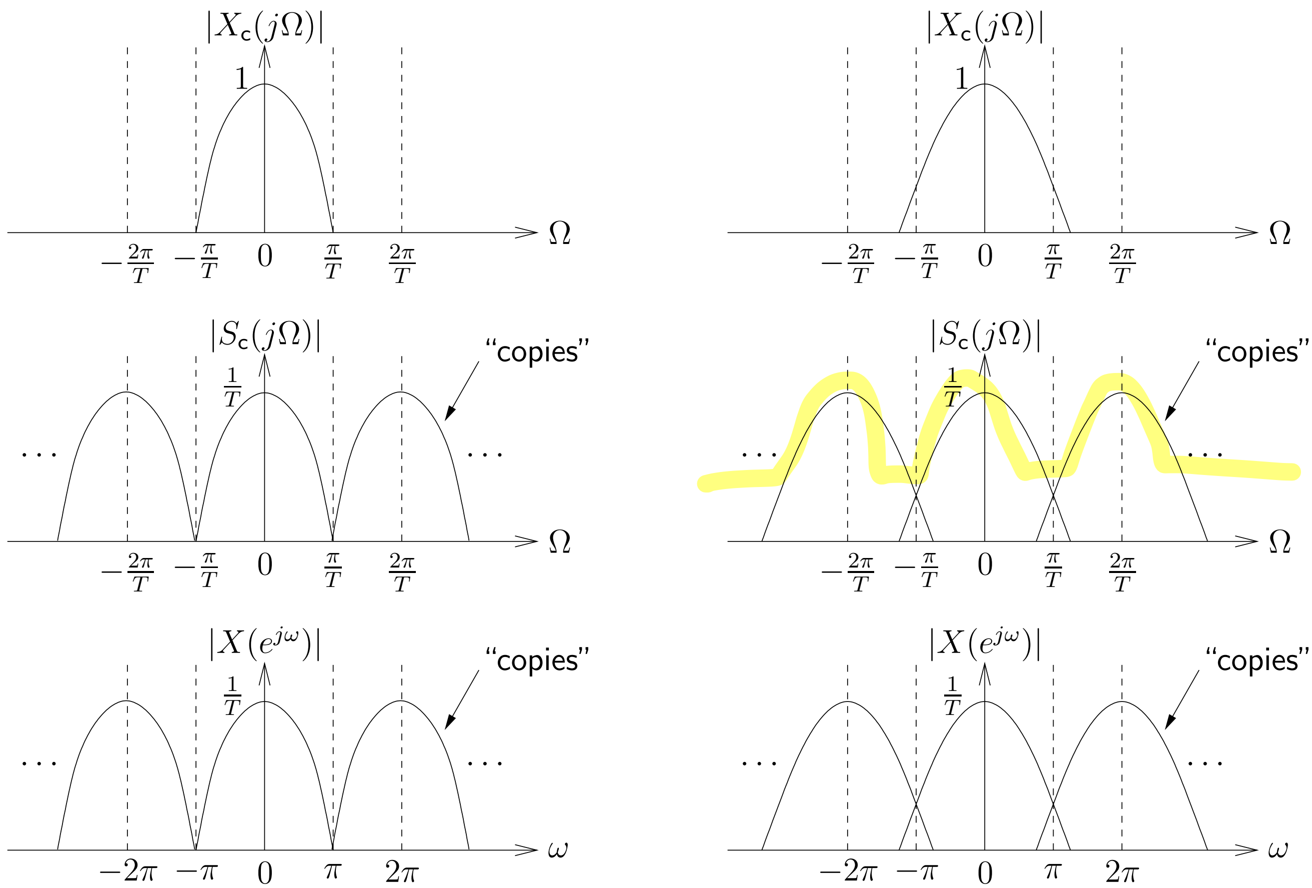
Finally, we can combine these two results into

$$X(e^{j\omega}) \;=\; S_c\big(j\tfrac{\omega}{T}\big) \;=\; \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c\big(j(\tfrac{\omega}{T} - \tfrac{2\pi}{T}k)\big)$$

or, equivalently, we could substitute $\boxed{\Omega \triangleq \frac{\omega}{T}}$ to write:

$$X(e^{j\Omega T}) \;=\; S_c\big(j\Omega\big) \;=\; \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c\big(j(\Omega - \tfrac{2\pi}{T}k)\big).$$

Both can be visualized as follows:



Sampling — DTFT/CTFT relationship (cont):

In words, the equation

$$\boxed{X(e^{j\omega}) \;=\; \frac{1}{T}\sum_{k=-\infty}^{\infty} X_{\mathsf{c}}\big(j(\tfrac{\omega-2\pi k}{T})\big)}$$

says that the DTFT spectrum $X(e^{j\omega})$ is composed of

- amplitude-scaled (by factor $\frac{1}{T}$),
- frequency-stretched (by factor T), and
- $2\pi k$-shifted

copies of $X_{\mathsf{c}}(j\Omega)$. These copies ensure 2π-periodicity.

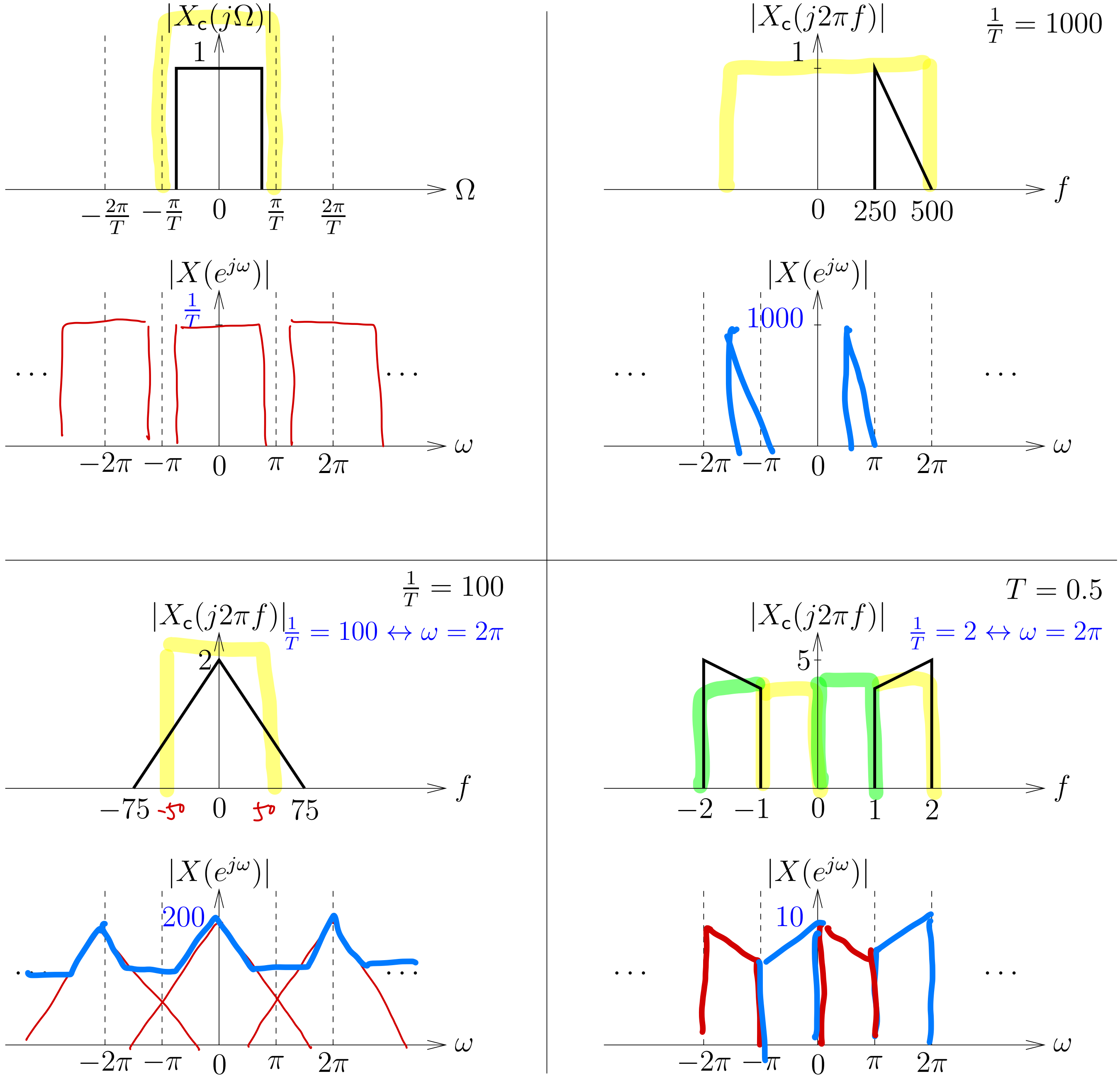
Note: The equation above gives a complete characterization of $\frac{1}{T}$-rate sampling, including the "aliasing" phenomenon seen earlier using sinusoidal examples.

Sampling — Examples:

$\Omega \in [-\frac{\pi}{T}, \frac{\pi}{T}]$

$f \in [-\frac{1}{2T}, \frac{1}{2T}]$

$T\Omega = \omega$



Remember:

$$\text{The sampling freq} \begin{cases} f = \frac{1}{T} \text{ Hz} \\ \Omega = \frac{2\pi}{T} \frac{\text{rad}}{\text{sec}} \end{cases} \text{ maps to } \omega = 2\pi \frac{\text{rad}}{\text{samp}}.$$